TEXORA · LECTURE SHEET

Hand Tools, Measuring Instruments and Carpentry Tools

The workshop's hands and eyes: cutting, striking and holding hand tools; marking out; precision measurement with the vernier caliper and micrometer — least count, reading, zero error — and the carpenter's kit for wood.

ME-1208 · ENGINEERING WORKSHOP · TOPIC 1 · LEVEL 1 TERM 2

1. Hand tools of the fitting bench

GROUP	TOOLS	KEY FACTS
Cutting	Hacksaw, files, chisels, scrapers, reamers	Hacksaw blades are graded by teeth per inch: coarse 14–18 TPI for thick soft stock, fine 24–32 for thin sections — keep at least 3 teeth in the cut. Files: single/double cut; grades bastard → second-cut → smooth → dead smooth; shapes flat, round, half-round, square, triangular
Striking	Ball-peen, cross-peen, straight-peen hammers; mallets (rawhide/wood — no surface damage)	Hammer mass is matched to the job; a mallet persuades without bruising
Holding & turning	Bench vice, V-blocks, clamps; spanners (open, ring, socket), adjustable wrench, pliers, screwdrivers	A ring spanner grips all six flats of a nut — less rounding than open-ended
Marking out	Scriber, centre & dot punches, dividers, odd-leg (jenny) caliper, surface plate, angle plate, marking blue	Mark from datum faces; centre-punch every drill point so the drill cannot wander

Care rules that examiners love: never use a file without a handle or as a lever; push the hacksaw on the forward stroke (teeth point forward); oil precision tools and store them separated — a workshop's accuracy dies by a thousand small knocks.

2. Precision measurement

The steel rule reads to about 0.5 mm. Beyond it stand the two workshop standards.

Vernier caliper

A sliding vernier scale subdivides the main scale. If 50 vernier divisions equal 49 main-scale divisions (1 MSD = 1 mm), the **least count** is

$$\text{LC} = 1\ \text{MSD} - 1\ \text{VSD} = 1 - 49/50 = 0.02\ \text{mm}$$

Reading = main-scale reading (just before the vernier zero) + (coinciding vernier line × LC). Example: main scale 23 mm, 12th line coincides → 23 + 12 × 0.02 = **23.24 mm**. The vernier measures outside, inside (upper jaws) and depth (the tail rod) in one instrument — the fitter's pocket in one tool.

Micrometer screw gauge

A precision screw advances the spindle: pitch 0.5 mm per turn, thimble divided into 50, so LC = 0.5/50 = **0.01 mm**. Reading = sleeve reading + thimble line × LC: sleeve 5.5 mm visible, thimble line 28 → 5.5 + 0.28 = **5.78 mm**. Close with the ratchet for a constant measuring force.

Zero error

With jaws closed the instrument must read zero. If it reads +0.03 mm, every measurement is 0.03 too big: **corrected = observed − zero error** (a negative error is added instead). Check before measuring, apply after — in that order.

Supporting cast: **vernier height gauge** for marking and heights on the surface plate; **dial indicator** (LC 0.01 mm) for run-out and alignment; **feeler gauges** for clearances; **thread pitch gauges** and **radius gauges** for profiles; **combination set** (rule, square head, centre head, protractor head) for layout work.

3. Carpentry tools

Wood is worked against and across its grain, and the tools say so: **rip saw** (coarse teeth, filed straight across) cuts along the grain; **crosscut saw** (finer, knife-edged teeth) severs across it; tenon and dovetail saws (backed by a stiffening strip) cut joints precisely. **Jack planes** (roughing) and **smoothing planes** flatten and finish; bevel-edged **chisels** (always paired with a mallet, never a steel hammer) pare joints; **firmer chisels** take heavier cuts; the **marking gauge** scribes lines parallel to an edge; **try square** and **mitre square** police right angles and 45°; braces and hand drills bore holes; rasps and spokeshaves shape curves. Rules: cut on the waste side of the line, plane with the grain, and treat a plane iron's sharpness as the whole craft in miniature.

4. Key points

- Hacksaw teeth point forward, cut on the push; pitch chosen so ≥ 3 teeth engage the section.
- Vernier LC = 1 MSD − 1 VSD (0.02 mm for the 49/50 pattern); micrometer LC = pitch ÷ thimble divisions (0.01 mm).
- Corrected reading = observed − zero error; sign discipline decides pass or fail.
- Marking out runs from datum faces on the surface plate, with punched centres for drilling.
- Carpentry: rip vs crosscut teeth, jack vs smoothing planes, chisels with mallets — and always cut on the waste side.

5. Practice

Answer in your own words; brief model answers follow.

1. A vernier caliper (LC 0.02 mm) shows a main-scale reading of 15 mm with the 8th vernier line coinciding. What is the measurement? The instrument has a zero error of +0.04 mm — give the corrected value.
2. A micrometer’s screw has a pitch of 0.5 mm and 50 thimble divisions. Derive its least count, then read: sleeve shows 7.5 mm, thimble line 23 coincides.
3. You must saw through (a) a 40 mm mild-steel bar and (b) a 1 mm steel sheet. Choose blade pitches and justify the (b) choice.
4. Why is a ring spanner preferred over an open-ended spanner for a tight, corroded nut?
5. A carpenter planes a board and the surface tears out in patches. Give the likely cause in terms of grain direction and the fix.
6. Name the instrument and procedure to check that a lathe shaft runs true (no wobble) without removing it from the centres.

Answer key

1. Measurement = 15 + 8 × 0.02 = 15.16 mm. Corrected = 15.16 − 0.04 = 15.12 mm. (Positive zero error means the instrument over-reads; subtract it.)
2. LC = 0.5/50 = 0.01 mm. Reading = 7.5 + 23 × 0.01 = 7.73 mm.
3. (a) Coarse blade, 14–18 TPI — thick section, big gullets clear the chips. (b) Fine blade, 32 TPI: the rule is at least three teeth in contact with the material; a 1 mm sheet under a coarse blade would have a single tooth straddling it, which snags, strips teeth and ruins the cut.
4. The ring surrounds the nut and bears on all six flats/corners, spreading the torque over the full hexagon; an open spanner contacts only two flats, concentrates the load on two corners and —

especially on a corroded, over-tight nut — rounds them off and slips. The ring also cannot slip off sideways.

5. The plane is travelling against the grain: where grain slopes up toward the cutter, the iron wedges under the fibres and lifts chunks out ahead of the edge — tear-out. Fix: reverse the board (plane with the grain), set the iron finer with the cap iron close, and skew the plane; stubborn patches yield to a scraper.
6. A dial indicator on a magnetic stand: tip riding on the shaft surface, shaft rotated slowly between centres; the needle’s total swing (total indicator reading) is the run-out. It measures truth of rotation to 0.01 mm without disturbing the setup.
